A UNIVERSAL RING FOR IDEMPOTENTS OF A FINITE FINITELY PRESENTED ALGEBRA

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Editorial note. Counted span is the author's own proof environment, source lines 22799-22832, immediately following the statement at 22790-22797.

Editorial note. Two cross-chapter pointers (Algebra) are recorded in assembly.external_references.

Editorial note. The universal ring construction is carried out explicitly with generators and equations inside the span.

Allowed standard prerequisites (editorial).

- Finiteness of an integral element. Algebra, lemma algebra-lemma-finite-is-integral, cited at proof line 22802. Produces a monic polynomial relation for each chosen generator of B over A.
- Composition of finite type ring maps. Algebra, lemma algebra-lemma-compose-finite-type, cited in the proof. Keeps the constructions of finite presentation while adjoining the universal idempotent data.
- Finite presentation of a quotient of a finite free algebra. used at proof lines 22803 onward. Presents B as a quotient of a finite free A-algebra, so that idempotents can be handled by polynomial equations with universal coefficients.

Source reference. Original reference 00F4: focused-seeds/geometry/S9-morph-open-closed-in-finite/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00GK: focused-seeds/geometry/S9-morph-open-closed-in-finite/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $A \to B$ be a ring map which is finite and of finite presentation. There exists a finitely presented ring map $A \to A_{univ}$ and an idempotent $e_{univ} \in B \otimes_A A_{univ}$ such that for any ring map $A \to A'$ and idempotent $e \in B \otimes_A A'$ there is a ring map $A_{univ} \to A'$ mapping e_{univ} to e.*

Proof. Choose $b_1, \ldots, b_n \in B$ generating B as an A-module. For each i choose a monic $P_i \in A[x]$ such that $P_i(b_i) = 0$ in B, see Algebra, Lemma 00GK. Thus B is a quotient of the finite free A-algebra $B' = A[x_1, \ldots, x_n]/(P_1(x_1), \ldots, P_n(x_n))$. Let $J \subset B'$ be the kernel of the surjection $B' \to B$. Then $J = (f_1, \ldots, f_m)$ is finitely generated as B is a finitely generated A-algebra, see Algebra, Lemma 00F4. Choose an A-basis $b'_1, \ldots, b'_N$ of B'. Consider the algebra

$$A_{univ} = A[z_1, \ldots, z_N, y_1, \ldots, y_m]/I$$

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where I is the ideal generated by the coefficients in $A[z_1, \ldots, z_n, y_1, \ldots, y_m]$ of the basis elements $b'_1, \ldots, b'_N$ of the expression

$$(\sum z_j b'_j)^2 - \sum z_j b'_j + \sum y_k f_k$$

in $B'[z_1, \ldots, z_N, y_1, \ldots, y_m]$. By construction the element $\sum z_j b'_j$ maps to an idempotent e_{univ} in the algebra $B \otimes_A A_{univ}$. Moreover, if $e \in B \otimes_A A'$ is an idempotent, then we can lift e to an element of the form $\sum b'_j \otimes a'_j$ in $B' \otimes_A A'$ and we can find $a''_k \in A'$ such that

$$(\sum b'_j \otimes a'_j)^2 - \sum b'_j \otimes a'_j + \sum f_k \otimes a''_k$$

is zero in $B' \otimes_A A'$. Hence we get an A-algebra map $A_{univ} \to A$ sending z_j to a'_j and y_k to a''_k mapping e_{univ} to e. This finishes the proof. $\square$